SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: **ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS** **COURSE CODE:** **MLA01**

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50 **Time: 60 Mins**

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I, D.Raghava(192525304) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty **Total Marks Awarded**

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	**Healthcare**	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	**Finance**	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	**Retail**	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	**Autonomous Vehicles**	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	**Robotics**	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	**Manufacturing**	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

Structure of the Report:

S. No	Report Component	Student Deliverable / Guiding Question
1	**Title of AI Application**	Provide the name of the latest AI application selected and the corresponding AI domain.
2	**Domain Selection**	Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing).
3	**Introduction and Background**	Describe the AI application and provide an overview of its purpose and importance.
4	**Problem Identification**	Explain the real-world problem or challenge addressed by this AI application.
5	**Need for AI Solution**	Analyze why AI is required and how it improves traditional approaches.
6	**AI Technologies Used**	Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).
7	**Working Mechanism**	Explain how the AI system works, including input data, processing methods, decision-making, and output generation.
8	**Data Sources and Processing**	Describe the type of data used and how the data is processed for intelligent decision-making.
9	**Real-World Implementation**	Identify organizations, industries, or real-world scenarios where this AI application is currently implemented.
10	**Impact and Benefits**	Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation.
11	**Challenges and Limitations**	Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.
12	**Future Scope and Emerging Trends**	Explain possible enhancements, future developments, and research opportunities in this AI application.
13	**Conclusion**	Summarize the significance of the AI application and its contribution to solving real-world problems.
14	**References**	Provide research papers, technical articles, industry reports, and reliable sources referred for the study.

1. Title of AI Application

AI-Supported Mammography Screening (the MASAI approach, using the Transparent AI system) for population breast cancer screening. AI domain: medical imaging, using deep learning and computer vision.

2. Domain Selection

Healthcare (Question 1: AI-based radiology and diagnosis). The application supports radiologists in reading screening mammograms for early breast cancer detection.

3. Introduction and Background

AI-supported mammography uses a trained deep learning model to read screening mammograms alongside a radiologist. The system looks at the images, marks suspicious regions, and gives each examination a score for how likely cancer is.

The clearest evidence comes from the Mammography Screening with Artificial Intelligence (MASAI) trial in Sweden. It is a randomised controlled trial of more than 105,000 women aged 40 to 74, in which each screening examination was randomly assigned to either AI-supported reading or standard double reading by two radiologists. The Transparant AI system was used in the AI-supported arm. The final results were published in The Lancet in early 2026, which makes this one of the most recent and strongest pieces of evidence for AI in clinical screening.

The application matters because breast cancer screening is a high-volume, repetitive task where small gains in detection save lives, and where many countries do not have enough radiologists.

4. Problem Identification

Organized screening programs in many countries use double reading: two radiologists independently read every mammogram. This catches more cancers than single reading but creates three problems.

- **Workforce shortage and workload.** Reading every image twice uses a large share of scarce radiologist time, and the screening-reading workload keeps growing as populations age and programs expand.
- **Missed and late-detected cancers.** Even with double reading, some cancers are missed and appear as interval cancers, meaning cancers diagnosed between two screening rounds. These tend to be more aggressive.
- **Unwanted recalls.** Women are sometimes called back for further tests that turn out to be normal, causing anxiety and extra cost.

The real-world challenge is therefore to detect more cancers earlier, without raising false alarms and without needing more radiologists.

5. Need for AI Solution

Traditional double reading depends entirely on human attention, which is limited by fatigue, volume and variation between readers. AI helps in three ways.

Aspect	Traditional double reading	AI-supported reading
Reader effort	Two radiologists read every exam	AI triages; most exams get one reader
Consistency	Varies between readers and over a shift	Same model scores every image the same way
Detection support	None beyond reader skill	Marks suspicious regions for the radiologist
Scalability	Limited by number of radiologists	Scales with computing capacity

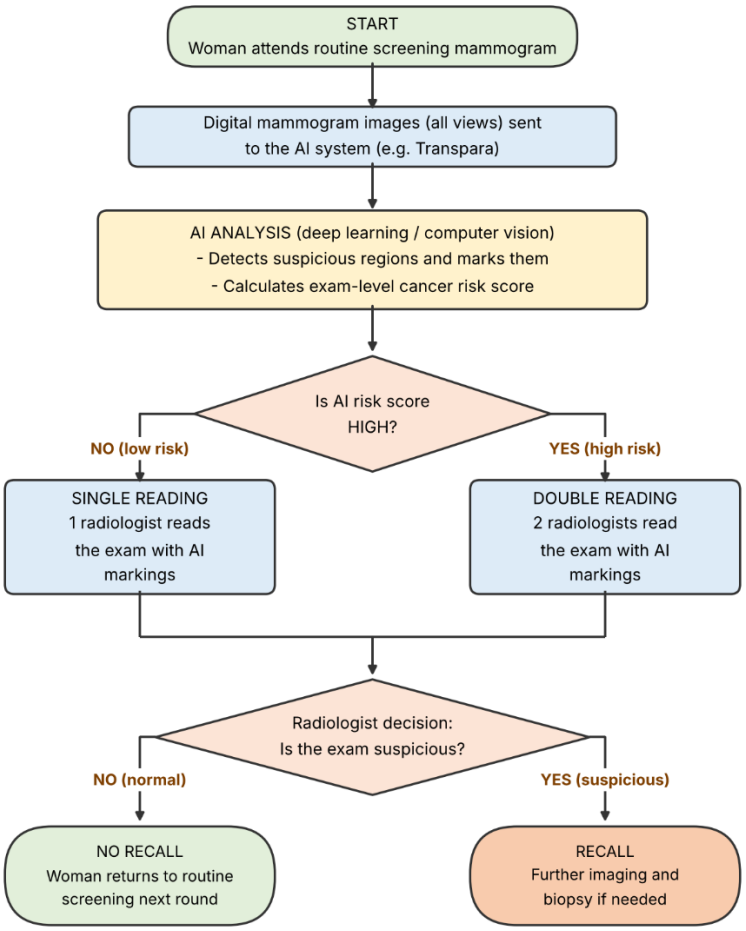
AI is not a replacement: the MASAI authors state that their results do not support replacing health-care professionals, because at least one radiologist still reads every examination with AI support.

6. AI Technologies Used

- **Deep learning (convolutional neural networks).** Networks trained on large sets of labelled mammograms learn visual patterns of malignancy such as masses, calcifications and architectural distortion.
- **Computer vision.** Image analysis locates and outlines suspicious regions on each view of the breast.
- **Supervised classification and risk scoring.** The model outputs a numeric exam-level score showing how suspicious the exam is. Related research systems such as Mirai predict future breast cancer risk directly from a mammogram.
- **Decision-support / triage logic.** Rule-based thresholds on the AI score decide whether an exam goes to single or double reading.
- **Human-in-the-loop design.** The radiologist makes the final decision; the AI only supports it.

These are classification and heuristic-style decision techniques applied to a real-world optimization problem: maximize detection and minimize workload.

7. Working Mechanism



Final decision is always made by the radiologist; the AI only supports and triages.

8. Data Sources and Processing

- **Training data.** Large archives of digital mammograms (2D, and increasingly 3D tomosynthesis) with confirmed outcomes: biopsy-proven cancers, normal follow-ups and interval cancers.
- **Operational data.** Each new screening exam, with its standard views of both breasts, arrives in DICOM image format from the hospital imaging system.
- **Processing.** Images are normalised, passed through the neural network, and converted into a score and region markings. Scores are compared with fixed thresholds for triage.
- **Validation data.** Performance is checked on data from different screening sites and populations. For example, Health New Zealand's planned validation includes mammograms from Māori, Pacific and Asian women to test fairness across groups.
- **Outcome data.** Registry follow-up (later diagnoses and interval cancers) is used to measure real-world accuracy.

9. Real-World Implementation

Where	What happened
Southwest Sweden (MASAI)	Randomised trial inside the Swedish screening programme, across four screening sites, with Transpara as the AI system (ASCO Post).
United States (ASSURE study)	Largest real-world US evaluation of AI-assisted screening; reported a 21.6% increase in cancer detection when AI was added to 3D mammography (Cleveland Clinic).
Urban safety-net hospital (USA)	Prospective deployment of the Mirai AI risk model to speed up follow-up after screening, published in npj Digital Medicine in May 2026 (Chung et al.).
New Zealand	Health NZ released a NZ$4.4 million, three-year tender in 2026 for AI reading of screening mammograms, starting with validation in July 2026 (eHealthNews).
Norway	AIMS Norway randomized trial protocol (2026) tests risk-stratified AI reading inside Breast Screen Norway (medRxiv).
Europe	Vendor Vara announced a CE mark for its tomosynthesis AI in June 2026, and reports Italy as the first European country to formally recommend AI in organised screening (Vara); this is a company claim and should be checked against official sources.

10. Impact and Benefits

In the MASAI trial, AI-supported screening did at least as well as double reading on the main safety measure and better on several others.

Measure	Standard double reading	AI-supported screening
Interval cancers per 1,000 women	1.76	1.55 (12% fewer; non-inferior)
Sensitivity	73.8%	80.5%
Specificity	98.5%	98.5%
Cancer detection (earlier analysis)	Baseline	29% higher, no rise in false positives
Screen-reading workload	Baseline	44% lower

Sources: Epocrates summary of the Lancet paper, AJMC, Applied Radiology.

- **Efficiency:** about 44% less reading work frees radiologists for diagnostic and other tasks.
- **Accuracy:** more cancers found, including fewer invasive interval cancers (16% fewer).

- **Cost and access:** the authors say cost-effectiveness still needs to be shown, but the workload saving is most valuable where radiologists are scarce.
- **Patient experience:** no rise in false positives, so no extra unnecessary recalls.

11. Challenges and Limitations

- **Overdiagnosis.** Both MASAI and the Korean AI-STREAM study found more in situ cancers with AI, which may add to overdiagnosis, since some of these cancers would never cause harm (PMC review).
- **Weaker on some cancers.** Sensitivity for in situ cancer was lower with AI support than in the control group (Wiley, Cancer).
- **Generalisability and bias.** Models may perform differently across populations, equipment and breast density; this is why New Zealand plans to validate on Māori, Pacific and Asian data (review).
- **Automation bias.** Radiologists might over-trust the AI. MASAI saw no change in specificity, which suggests this did not happen there.
- **Evidence gaps.** Cost-effectiveness and effects across later screening rounds are not yet known, and MASAI is one country's programme with one vendor's system.
- **Regulation, privacy and accountability.** Regulatory approval, patient-data protection and who is responsible when AI and radiologist disagree remain open issues.
- **Integration.** The AI must fit existing hospital IT systems and screening workflows.

12. Future Scope and Emerging Trends

- **Risk-based, personalized screening.** AI risk models such as Mirai could set screening intervals and extra imaging for each woman instead of one rule for everyone.
- **3D mammography (tomosynthesis).** AI for tomosynthesis is gaining regulatory approval in Europe and showed gains in the US ASSURE study.
- **More trials.** Norway's AIMS trial and other studies will test other triage strategies and populations.
- **National programs.** Countries such as New Zealand are moving from validation to production use.
- **Research opportunities.** Explainable AI, fairness across ethnic groups and breast density, cost-effectiveness studies, and AI that learns safely from new data after deployment.

13. Conclusion

AI-supported mammography shows that AI can safely share the work of screening with radiologists. In the largest randomized trial so far, one radiologist plus AI found more cancers than two radiologists and cut reading workload by about 44%, without more false alarms. Open issues remain: overdiagnosis, fairness across populations, cost-effectiveness and regulation. Still, the evidence is strong enough that several countries are now moving to adopt it, making it a good example of AI solving a real healthcare problem while keeping humans in control.

14. References

1. Gommers J, et al. Interval cancer, sensitivity, and specificity comparing AI-supported mammography screening with standard double reading without AI in the MASAI study. The Lancet, 2026. Epocrates summary; ScienceDirect
2. ClinicalTrials.gov. Mammography Screening With Artificial Intelligence (MASAI), NCT04838756. Link
3. The ASCO Post. Randomized Trial Shows AI-Supported Mammography Improves Sensitivity and Lowers Interval Cancer Rate, Feb 2026. Link
4. AJMC. AI-Supported Mammography Caught More Cancers During Screening. Link
5. Applied Radiology. MASAI Trial Demonstrates AI-Assisted Mammography Screening Increases Cancer Detection, Decreases Radiologist Workload. Link
6. Lawrence. MASAI trial results support larger scale use of AI-supported mammography. Cancer, 2026. Link
7. Artificial Intelligence for Mammography Ready for the Real-World. PMC. Link
8. Chung M, et al. Prospective deployment of AI-based risk stratification to enable expedited mammography workflow in a safety-net setting. npj Digital Medicine, May 2026. Link
9. AIMS Norway: Protocol for a randomized controlled trial. medRxiv, 2026. Link
10. Health NZ to invest $4.4M in AI solutions for breast screening. eHealthNews.nz, May 2026. Link
11. Cleveland Clinic. Breast cancer screening moves toward a more personalized future, Aug 2026. Link
12. Artificial Intelligence in Mammography Screening: A Narrative Review of Progress, Pitfalls, and Potential. British Journal of Radiology, 2026. Link

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement (
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI applicatio n.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanatio n with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplet e.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanatio n with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriat e citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementa t ion analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluatio n with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.

			evaluation.			
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.